

Forecasting Exchange-Rate Volatility with MEM and HAR Models

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Introduction

The aim of this analysis is to forecast the daily volatility of the exchange rates of several cryptocurrencies and fiat currencies against the US dollar.

Volatility is forecast using MEM(1,1), MEM(2,1), and HAR models, with the objective of identifying the best-performing model. The models are subsequently compared using forecast-error measures and Diebold–Mariano tests.

Chapter 1

Data and Methodology

1.1 Data

The data analysed are daily exchange rates against the US dollar for two currency categories: fiat currencies (left panel of Table 1.1) and cryptocurrencies (right panel). Fiat currencies, Bitcoin Cash (BCH), and Litecoin (LTC) are considered over the period from 17 September 2014 to 1 February 2024; the remaining cryptocurrencies are considered from 9 September 2017 to 1 February 2024.

The data were obtained from Yahoo Finance (<https://finance.yahoo.com>). The Chinese yuan (CNY), UAE dirham (AED), and Thai baht (THB) exchange rates were initially included but were subsequently excluded because anomalies were found in the data.

Table 1.1: Fiat and cryptocurrency exchange rates.

Fiat currencies		Cryptocurrencies	
Symbol	Name	Symbol	Name
EUR	Euro	BTC	Bitcoin
JPY	Japanese yen	LTC	Litecoin
GBP	Pound sterling	ETH	Ether
AUD	Australian dollar	ADA	Cardano
CAD	Canadian dollar	XRP	XRP
CHF	Swiss franc	XLM	Stellar
KRW	South Korean won	BCH	Bitcoin Cash
TWD	New Taiwan dollar	DOGE	Dogecoin
BRL	Brazilian real	ETC	Ethereum Classic
TRY	Turkish lira	TRX	Tron
RUB	Russian ruble	LINK	Chainlink
INR	Indian rupee	BNB	Binance Coin

1.1.1 Volatility

Because daily volatility σ_t is not directly observable, it is typically measured using a proxy. The proxy employed in this analysis is the high–low range proposed by Garman and Klass (1980):

$$GK_t = \frac{1}{2} \left[\ln \left(\frac{H_t}{L_t} \right) \right]^2 - [2 \ln(2) - 1] \left[\ln \left(\frac{C_t}{O_t} \right) \right]^2. \quad (1.1)$$

Here O_t is the opening price on day t , H_t is the highest price on day t , L_t is the lowest price on day t , and C_t is the closing price on day t . Note that $GK_t \geq 0$ for every t .

1.2 Rolling Analysis

The models were compared using a rolling-window approach (Hyndman, 2016, *Cross-validation for time series*). This approach uses moving time windows, that is, intervals of past observations used to generate future forecasts. Both the window length and the forecast interval—the frequency with which forecasts are produced—must be specified.

Starting from the beginning of the time series $\{y_t\}$, an interval of observations is selected and used to forecast the following period. Once the forecast has been produced, the data window moves forward by one period and another one-step-ahead forecast is computed. This procedure continues until all available observations have been used.

1.3 Models

1.3.1 MEM

Let $\{y_t\}$ be a process defined on $[0, +\infty)$, $t \in \mathbb{N}$, and let I_{t-1} denote the information available for forecasting y_t . The process follows a Multiplicative Error Model (MEM; Brownlees, Cipollini, and Gallo, 2011) if it can be written as

$$y_t = \mu_t \varepsilon_t, \quad (1.2)$$

where, conditionally on I_{t-1} , μ_t is a positive quantity that evolves deterministically according to a parameter vector $\boldsymbol{\theta}$,

$$\mu_t = \mu(\boldsymbol{\theta}, I_{t-1}), \quad (1.3)$$

and ε_t is a random variable with a probability density supported on $[0, +\infty)$, unit mean, and constant variance:

$$\varepsilon_t \mid \mathcal{F}_{t-1} \sim D^+(1, \sigma^2). \quad (1.4)$$

Equations (2)–(4) imply

$$E(y_t \mid I_{t-1}) = \mu_t, \quad (1.5)$$

$$V(y_t \mid I_{t-1}) = \sigma^2 \mu_t^2. \quad (1.6)$$

Including lagged returns r_{t-1} in I_{t-1} and defining $y_t^{(-)} \equiv y_t \mathbb{I}(r_t < 0)$, where $\mathbb{I}(\cdot)$ is the indicator function, the reference specification for μ_t is

$$\mu_t = \omega + \beta_1 \mu_{t-1} + \alpha_1 y_{t-1} + \gamma_1 y_{t-1}^{(-)} + \alpha_2 y_{t-2}, \quad (1.7)$$

which is labelled MEM(2,1). Setting $\alpha_2 = 0$ gives

$$\mu_t = \omega + \beta_1 \mu_{t-1} + \alpha_1 y_{t-1} + \gamma_1 y_{t-1}^{(-)}, \quad (1.8)$$

which is labelled MEM(1,1).

If $\{y_t\}$ is mean stationary, then $E(y_t) = E(\mu_t) = \mu$. Taking expectations on both sides of (7) gives

$$\omega = \mu - \left(\beta_1 + \alpha_1 + \alpha_2 + \frac{\gamma_1}{2} \right) \mu. \quad (1.9)$$

The distribution of the error term ε_t may be specified using any probability density satisfying the properties stated in (4).

1.3.2 HAR

Define

$$y_{t-1}^{(rw)} = \frac{1}{4} \sum_{j=2}^5 y_{t-j}, \quad (1.10)$$

$$y_{t-1}^{(rm)} = \frac{1}{17} \sum_{j=6}^{22} y_{t-j}. \quad (1.11)$$

The HAR model (Corsi, 2009) is

$$y_t = \beta_0 + \beta_1 y_{t-1} + \beta_w y_{t-1}^{(rw)} + \beta_m y_{t-1}^{(rm)} + \gamma_1 y_{t-1}^{(-)} + \varepsilon_t, \quad (1.12)$$

where ε_t is a zero-mean random variable. Relative to the usual specification, the asymmetric effect $\gamma_1 y_{t-1}^{(-)}$ is added to ensure comparability with the MEM (Tauchen, *The AR and HAR Model*).

1.4 Evaluation of Results

The forecasts produced by the models described above were compared using forecast-error measures and Diebold–Mariano tests.

1.4.1 Forecast-error measures

The forecast-error measures provide a quantitative assessment of the predictive accuracy of each model. The measures used here are Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), Root Mean Squared Percentage Error (RMSPE), and Log-Likelihood Error (LLE). Let y_t be the observed values and $\hat{y}_t$ the values forecast by a model. The measures are calculated as

$$\text{MAE} = \frac{1}{T} \sum_{t=1}^T |y_t - \hat{y}_t|, \quad (1.13)$$

$$\text{RMSE} = \sqrt{\frac{1}{T} \sum_{t=1}^T (y_t - \hat{y}_t)^2}, \quad (1.14)$$

$$\text{MAPE} = \frac{1}{T} \sum_{t=1}^T \left| \frac{y_t - \hat{y}_t}{y_t} \right|, \quad (1.15)$$

$$\text{RMSPE} = \sqrt{\frac{1}{T} \sum_{t=1}^T \left(\frac{y_t - \hat{y}_t}{y_t} \right)^2}, \quad (1.16)$$

$$\text{LLE} = \frac{1}{T} \sum_{t=1}^T \left[\frac{y_t}{\hat{y}_t} - 1 - \ln \left(\frac{y_t}{\hat{y}_t} \right) \right]. \quad (1.17)$$

1.4.2 Diebold–Mariano test

The Diebold–Mariano test (Diebold and Mariano, 1995) determines whether the forecast-error measures of two models differ significantly. When more than two models are considered, pairwise comparisons are performed.

Let $g(y_t, \hat{y}_t) \geq 0$ be a loss function such as AE, SE, or LLE:

$$SE_t = (y_t - \hat{y}_t)^2, \quad (1.18)$$

$$LLE_t = \frac{y_t}{\hat{y}_t} - 1 - \ln \left(\frac{y_t}{\hat{y}_t} \right), \quad (1.19)$$

$$AE_t = |y_t - \hat{y}_t|. \quad (1.20)$$

The difference between two losses is

$$d_t = g(y_t, \hat{y}_{1t}) - g(y_t, \hat{y}_{2t}). \quad (1.21)$$

Assume henceforth that $\{d_t\}$ is weakly stationary and ergodic. The two sets of forecasts have the same average accuracy if and only if the mean loss difference is zero for all t . We therefore test

$$H_0 : E(d_t) = 0 \quad (1.22)$$

against one of the alternatives $E(d_t) < 0$, $E(d_t) \neq 0$, or $E(d_t) > 0$. The null states that the models have similar predictive accuracy. The alternatives respectively imply that the first model is better, the models differ, or the second model is better.

Under H_0 , the test statistic satisfies

$$\frac{\bar{d}}{\sqrt{V(\bar{d})}} \xrightarrow{d} N(0, 1), \quad (1.23)$$

where

$$\bar{d} = \frac{1}{T} \sum_{t=1}^T d_t, \quad (1.24)$$

$$V(\bar{d}) = \frac{1}{T} \left(\gamma_0 + 2 \sum_{n=1}^{\infty} \gamma_n \right), \quad (1.25)$$

and γ_n is the autocovariance function of $\{d_t\}$. It is typically estimated by

$$\hat{V}(\bar{d}) = \frac{1}{T} \left(\hat{\gamma}_0 + 2 \sum_{n=1}^M \hat{\gamma}_n \right), \quad (1.26)$$

where $M = \lfloor T^{1/3} \rfloor$ and $\hat{\gamma}_n$ is the sample estimate of γ_n (Diebold, 2015).

Chapter 2

Results

Figure 2.1 reports the series and autocorrelation function for the Litecoin (LTC) cryptocurrency exchange rate; Figure 2.2 gives the corresponding plots for the Japanese yen (JPY) fiat exchange rate.

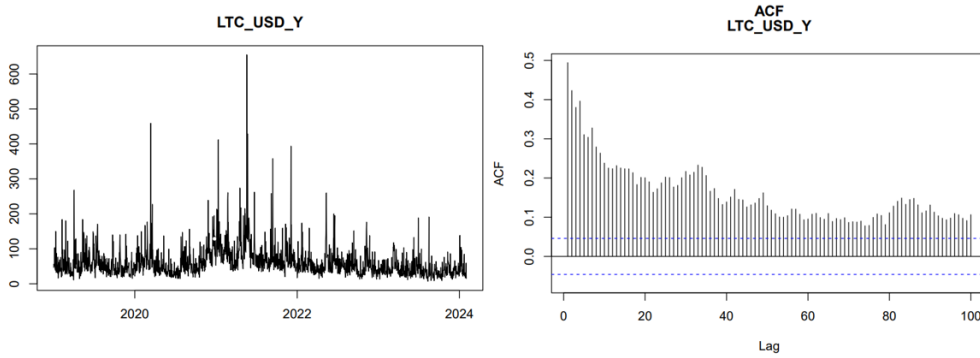


Figure 2.1: Series and ACF for the Litecoin (LTC) cryptocurrency.

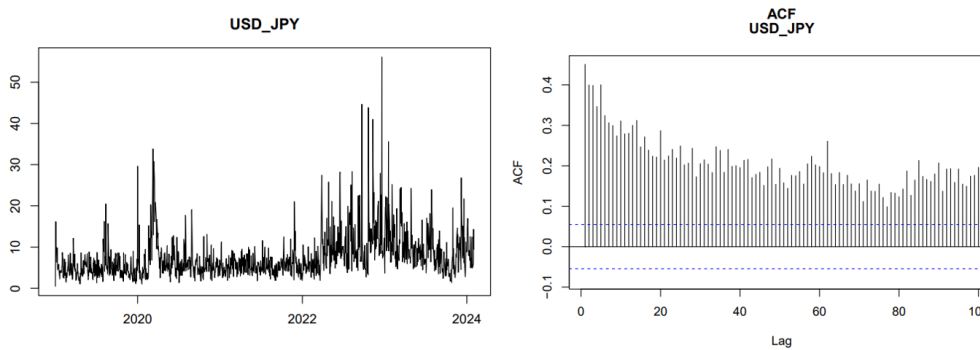


Figure 2.2: Series and ACF for the Japanese yen (JPY) fiat currency.

The estimated coefficients for Litecoin (LTC) and the euro (EUR) are reported for MEM(1,1), MEM(2,1), and HAR in Tables 2.1–2.3.

Table 2.1: Estimated MEM(1,1) coefficients for 01/01/2014–12/31/2018 and 01/01/2019–12/31/2019.

Symbol	Coefficient	Estimate	SE	z statistic
LTC	ω	3.335	0.6699	4.979
	β_1	0.622	0.0373	16.699
	α_1	0.340	0.0385	8.847
	γ_1	-0.035	0.0351	-1.001
	σ	0.820		
EUR	ω	0.066	0.0344	1.914
	β_1	0.916	0.0145	63.220
	α_1	0.082	0.0137	6.020
	γ_1	-0.015	0.0106	-1.409
	σ	0.470		

Table 2.2: Estimated MEM(2,1) coefficients for 01/01/2014–12/31/2018 and 01/01/2019–12/31/2019.

Symbol	Coefficient	Estimate	SE	z statistic
LTC	ω	0.828	0.2830	2.925
	β_1	0.881	0.0263	33.489
	α_1	0.459	0.0434	10.562
	α_2	-0.336	0.0506	-6.639
	γ_1	-0.035	0.0195	-1.766
	σ	0.810		
EUR	ω	0.065	0.0348	1.876
	β_1	0.917	0.0158	58.236
	α_1	0.086	0.0309	2.777
	α_2	-0.004	0.0334	-0.122
	γ_1	-0.015	0.0106	-1.414
	σ	0.470		

Table 2.3: Estimated HAR coefficients for 01/01/2014–12/31/2018 and 01/01/2019–12/31/2019.

Symbol	Coefficient	Estimate	SE	z statistic
LTC	β_0	7.779	2.1274	3.657
	β_1	0.481	0.0286	16.789
	β_w	0.196	0.0338	5.788
	β_m	0.196	0.0373	5.247
	γ_1	-0.004	0.0274	-0.150
	σ	45.110		
EUR	β_0	0.065	0.0348	1.876
	β_1	0.917	0.0158	58.236
	β_w	0.086	0.0309	2.777
	β_m	-0.004	0.0334	-0.122
	γ_1	-0.015	0.0106	-1.414
	σ	0.470		

2.1 Forecast-error measures

This section reports forecast-error measures for 12 cryptocurrency and 12 fiat exchange rates. Each cell (i, j) indicates how many times model i is best according to measure j .

2.1.1 Cryptocurrency exchange rates

The cryptocurrency results in Table 2.4 clearly show that MEM(2,1) is the best model for every measure and forecast horizon. HAR is best for only one cryptocurrency and only for MAE, RMSE, LLE, ScMAE, and ScRMSE.

Table 2.4: Forecast-error measures for cryptocurrency exchange rates.

h	Model	MAE	RMSE	MAPE	RMSPE	LLE
1	MEM(2,1)	11	12	12	12	12
	MEM(1,1)	0	0	0	0	0
	HAR	1	0	0	0	0
2	MEM(2,1)	11	11	12	12	12
	MEM(1,1)	0	0	0	0	0
	HAR	1	1	0	0	0
3	MEM(2,1)	11	11	12	12	11
	MEM(1,1)	0	1	0	0	0
	HAR	1	0	0	0	1
4	MEM(2,1)	11	11	12	12	10
	MEM(1,1)	0	0	0	0	0
	HAR	1	1	0	0	2
5	MEM(2,1)	11	11	12	12	10
	MEM(1,1)	0	0	0	0	1
	HAR	1	1	0	0	1

2.1.2 Fiat exchange rates

The fiat-currency results in Table 2.5 are much less clear-cut than the cryptocurrency results. Overall, MEM(1,1) is the best model.

Table 2.5: Forecast-error measures for fiat exchange rates.

h	Model	MAE	RMSE	MAPE	RMSPE	LLE
1	MEM(2,1)	3	1	4	6	4
	MEM(1,1)	5	7	3	3	6
	HAR	4	4	5	3	2
2	MEM(2,1)	1	2	1	1	3
	MEM(1,1)	8	7	7	7	6
	HAR	3	3	4	4	3
3	MEM(2,1)	1	3	1	2	4
	MEM(1,1)	8	6	7	6	3
	HAR	3	3	4	4	5
4	MEM(2,1)	2	3	2	3	4
	MEM(1,1)	7	6	7	5	5
	HAR	3	3	3	4	3
5	MEM(2,1)	4	2	2	4	4
	MEM(1,1)	5	7	7	5	6
	HAR	3	3	3	3	2

2.2 Diebold–Mariano Test

Each cell (i, j) in the following tables indicates how many times the row model is significantly better in predictive accuracy than the column model at $\alpha = 0.05$.

2.2.1 Cryptocurrency exchange rates

For cryptocurrencies, Diebold–Mariano tests based on SE at every forecast horizon (left panel of Table 2.6) indicate an advantage of MEM(2,1) over HAR and MEM(1,1). The LLE-based tests (right panel) confirm and strengthen this result.

Table 2.6: Diebold–Mariano tests for cryptocurrency exchange rates using MEM(2,1), MEM(1,1), and HAR; $\text{cell}(i, j) + \text{cell}(j, i) \leq 12$.

h	Model	Loss: SE			Loss: LLE		
		MEM(2,1)	MEM(1,1)	HAR	MEM(2,1)	MEM(1,1)	HAR
1	MEM(2,1)	–	1	7	–	10	7
	MEM(1,1)	0	–	0	0	–	0
	HAR	0	0	–	0	2	–
2	MEM(2,1)	–	3	4	–	9	5
	MEM(1,1)	0	–	0	0	–	0
	HAR	0	3	–	0	3	–
3	MEM(2,1)	–	5	4	–	9	6
	MEM(1,1)	0	–	0	0	–	0
	HAR	0	3	–	0	3	–
4	MEM(2,1)	–	7	4	–	10	4
	MEM(1,1)	0	–	0	0	–	0
	HAR	0	3	–	0	4	–
5	MEM(2,1)	–	10	6	–	10	7
	MEM(1,1)	0	–	0	0	–	0
	HAR	0	4	–	0	2	–

2.2.2 Fiat exchange rates

For fiat currencies, the SE-based tests indicate that MEM(1,1) and MEM(2,1) have no disadvantage relative to HAR at any forecast horizon. Between the two MEM specifications, the test shows a significant advantage for MEM(1,1), which has fewer disadvantages than MEM(2,1) at every horizon. The LLE-based tests reinforce this finding. Among the models examined, MEM(1,1) is the most effective, although the results are less clear-cut and the differences are smaller than for cryptocurrencies.

Table 2.7: Diebold–Mariano tests for fiat exchange rates using MEM(2,1), MEM(1,1), and HAR; $\text{cell}(i, j) + \text{cell}(j, i) \leq 12$.

h	Model	Loss: SE			Loss: LLE		
		MEM(2,1)	MEM(1,1)	HAR	MEM(2,1)	MEM(1,1)	HAR
1	MEM(2,1)	–	0	2	–	1	3
	MEM(1,1)	1	–	4	1	–	4
	HAR	0	0	–	0	0	–
2	MEM(2,1)	–	0	3	–	0	2
	MEM(1,1)	5	–	4	4	–	6
	HAR	0	0	–	0	0	–
3	MEM(2,1)	–	0	3	–	1	3
	MEM(1,1)	1	–	2	2	–	3
	HAR	0	0	–	0	0	–
4	MEM(2,1)	–	0	1	–	0	3
	MEM(1,1)	1	–	2	2	–	3
	HAR	0	0	–	0	0	–
5	MEM(2,1)	–	0	1	–	0	3
	MEM(1,1)	1	–	1	2	–	3
	HAR	0	0	–	0	0	–

Chapter 3

Conclusion

The analysis provided a detailed comparison of the forecasting ability, at different horizons, of several models for the daily volatility of exchange rates.

For cryptocurrencies, both the forecast-error measures and the Diebold–Mariano tests show that MEM(2,1) is the most effective model, except for Bitcoin (BTC), for which HAR and MEM(2,1) are essentially tied.

For fiat currencies, the results are less clear, although MEM(1,1) shows good relative performance.

Appendix A

Detailed Forecast-Error Tables

Table A.1: Forecast-error measures for the BTC exchange rate under the MEM and HAR models, rounded to four decimal places.

h	Model	Bitcoin (BTC)						
		MAE	RMSE	MAPE	RMSPE	LLE	ScMAE	ScRMSE
1	MEM(2,1)	17.0771	26.5393	0.5176	0.7793	0.1380	0.7428	0.7968
	MEM(1,1)	17.3329	26.7744	0.5284	0.7946	0.1410	0.7539	0.8038
	HAR	17.0285	26.5468	0.5221	0.7975	0.1392	0.7407	0.7970
2	MEM(2,1)	17.9811	28.3106	0.5618	0.8421	0.1583	0.7821	0.8499
	MEM(1,1)	18.3843	28.4904	0.5701	0.8580	0.1599	0.7997	0.8553
	HAR	17.9500	28.2811	0.5725	0.8672	0.1594	0.7808	0.8491
3	MEM(2,1)	18.4739	28.8850	0.5814	0.8702	0.1637	0.8036	0.8672
	MEM(1,1)	18.9506	29.0331	0.5944	0.8909	0.1644	0.8243	0.8716
	HAR	18.4475	28.7982	0.5931	0.8935	0.1637	0.8024	0.8646
4	MEM(2,1)	18.8043	29.3589	0.5932	0.8893	0.1673	0.8179	0.8814
	MEM(1,1)	19.2436	29.5660	0.6096	0.9174	0.1667	0.8370	0.8876
	HAR	18.7448	29.2013	0.6055	0.9123	0.1671	0.8153	0.8767
5	MEM(2,1)	19.1540	29.9679	0.6081	0.9165	0.1737	0.8331	0.8997
	MEM(1,1)	19.6893	30.3291	0.6301	0.9539	0.1735	0.8564	0.9105
	HAR	18.9981	29.8537	0.6192	0.9429	0.1736	0.8264	0.8963

Table A.2: Forecast-error measures for the LTC exchange rate under the MEM and HAR models, rounded to four decimal places.

h	Model	Litecoin (LTC)						
		MAE	RMSE	MAPE	RMSPE	LLE	ScMAE	ScRMSE
1	MEM(2,1)	23.5165	37.2991	0.4533	0.6291	0.1145	0.7422	0.8018
	MEM(1,1)	23.6164	37.3074	0.4583	0.6361	0.1158	0.7453	0.8020
	HAR	23.7517	37.6910	0.4583	0.6402	0.1160	0.7496	0.8102
2	MEM(2,1)	24.5017	38.7156	0.4872	0.6801	0.1281	0.7733	0.8323
	MEM(1,1)	24.9180	39.1435	0.4915	0.6894	0.1304	0.7864	0.8415
	HAR	24.5697	38.9183	0.4914	0.6906	0.1291	0.7754	0.8366
3	MEM(2,1)	25.2594	39.5643	0.5070	0.7100	0.1333	0.7972	0.8505
	MEM(1,1)	25.8756	40.1278	0.5159	0.7265	0.1356	0.8166	0.8626
	HAR	25.3862	39.8472	0.5112	0.7182	0.1340	0.8012	0.8566
4	MEM(2,1)	25.7594	40.0278	0.5208	0.7320	0.1369	0.8130	0.8605
	MEM(1,1)	26.4575	40.6816	0.5352	0.7571	0.1401	0.8350	0.8745
	HAR	25.9189	40.2771	0.5258	0.7395	0.1378	0.8180	0.8658
5	MEM(2,1)	26.3114	40.7711	0.5354	0.7558	0.1416	0.8304	0.8765
	MEM(1,1)	27.3216	41.8627	0.5577	0.7934	0.1470	0.8623	0.8999
	HAR	26.4928	41.2799	0.5408	0.7663	0.1437	0.8361	0.8874

Table A.3: Forecast-error measures for the ADA exchange rate under the MEM and HAR models, rounded to four decimal places.

h	Model	Cardano (ADA)						
		MAE	RMSE	MAPE	RMSPE	LLE	ScMAE	ScRMSE
1	MEM(2,1)	27.1926	42.6744	0.4679	0.6484	0.1175	0.6897	0.7781
	MEM(1,1)	27.9211	42.7866	0.4954	0.6891	0.1211	0.7082	0.7801
	HAR	27.8184	42.8849	0.4870	0.6797	0.1192	0.7056	0.7819
2	MEM(2,1)	29.6469	45.8329	0.5283	0.7352	0.1397	0.7520	0.8356
	MEM(1,1)	30.6798	46.1751	0.5631	0.7896	0.1441	0.7782	0.8419
	HAR	30.2157	45.9812	0.5509	0.7754	0.1416	0.7664	0.8383
3	MEM(2,1)	30.8360	47.3529	0.5575	0.7807	0.1486	0.7821	0.8634
	MEM(1,1)	32.0557	47.7711	0.6062	0.8623	0.1538	0.8131	0.8710
	HAR	31.5603	47.6599	0.5838	0.8264	0.1509	0.8005	0.8690
4	MEM(2,1)	31.2479	47.9436	0.5706	0.8023	0.1535	0.7926	0.8741
	MEM(1,1)	32.6836	48.5139	0.6340	0.9093	0.1603	0.8290	0.8845
	HAR	32.1268	48.2842	0.6018	0.8553	0.1562	0.8149	0.8803
5	MEM(2,1)	32.0541	49.1979	0.5890	0.8303	0.1611	0.8130	0.8970
	MEM(1,1)	33.8875	50.2191	0.6673	0.9614	0.1712	0.8595	0.9156
	HAR	33.0452	49.5587	0.6254	0.8937	0.1640	0.8382	0.9036

Table A.4: Forecast-error measures for the ETH exchange rate under the MEM and HAR models, rounded to four decimal places.

h	Model	Ether (ETH)						
		MAE	RMSE	MAPE	RMSPE	LLE	ScMAE	ScRMSE
1	MEM(2,1)	20.9234	33.3720	0.4778	0.7081	0.1180	0.7016	0.7663
	MEM(1,1)	21.2406	33.5509	0.4906	0.7344	0.1201	0.7122	0.7704
	HAR	21.2044	33.5641	0.4896	0.7335	0.1195	0.7110	0.7707
2	MEM(2,1)	22.1836	35.7139	0.5230	0.7928	0.1334	0.7438	0.8200
	MEM(1,1)	22.5324	35.8004	0.5345	0.8171	0.1356	0.7555	0.8220
	HAR	22.4588	35.8010	0.5429	0.8362	0.1360	0.7530	0.8220
3	MEM(2,1)	22.6409	36.7769	0.5419	0.8316	0.1410	0.7591	0.8445
	MEM(1,1)	23.0734	36.9345	0.5585	0.8660	0.1438	0.7736	0.8481
	HAR	22.9997	36.9208	0.5647	0.8781	0.1440	0.7712	0.8478
4	MEM(2,1)	23.0994	37.3865	0.5579	0.8611	0.1456	0.7745	0.8585
	MEM(1,1)	23.5251	37.6201	0.5786	0.9041	0.1493	0.7888	0.8638
	HAR	23.5008	37.5084	0.5826	0.9109	0.1490	0.7880	0.8613
5	MEM(2,1)	23.5633	38.2553	0.5738	0.8863	0.1520	0.7901	0.8784
	MEM(1,1)	24.0793	38.6266	0.5988	0.9359	0.1566	0.8074	0.8869
	HAR	24.0220	38.4681	0.6014	0.9435	0.1561	0.8054	0.8833

Table A.5: Forecast-error measures for the XRP exchange rate under the MEM and HAR models, rounded to four decimal places.

h	Model	Ripple (XRP)						
		MAE	RMSE	MAPE	RMSPE	LLE	ScMAE	ScRMSE
1	MEM(2,1)	28.0461	50.0201	0.5239	0.7672	0.1498	0.6715	0.7621
	MEM(1,1)	28.6714	50.5073	0.5440	0.7935	0.1538	0.6864	0.7695
	HAR	28.4901	51.0040	0.5331	0.7854	0.1524	0.6821	0.7770
2	MEM(2,1)	30.3580	53.6256	0.5894	0.8707	0.1742	0.7268	0.8170
	MEM(1,1)	31.4175	54.3064	0.6139	0.9050	0.1793	0.7522	0.8274
	HAR	30.8567	54.1240	0.6032	0.8895	0.1764	0.7388	0.8246
3	MEM(2,1)	31.3680	55.6203	0.6168	0.9146	0.1867	0.7510	0.8474
	MEM(1,1)	32.8137	56.6961	0.6555	0.9737	0.1942	0.7856	0.8638
	HAR	32.0334	56.0929	0.6342	0.9361	0.1894	0.7669	0.8546
4	MEM(2,1)	32.2489	56.7793	0.6341	0.9448	0.1950	0.7721	0.8650
	MEM(1,1)	33.7309	58.2457	0.6876	1.0364	0.2072	0.8076	0.8874
	HAR	32.8619	57.3563	0.6544	0.9725	0.1986	0.7868	0.8738
5	MEM(2,1)	32.7807	57.5660	0.6483	0.9700	0.2013	0.7848	0.8770
	MEM(1,1)	34.5895	59.3459	0.7199	1.0910	0.2178	0.8281	0.9041
	HAR	33.5556	58.2684	0.6751	1.0097	0.2059	0.8034	0.8877

Table A.6: Forecast-error measures for the XLM exchange rate under the MEM and HAR models, rounded to four decimal places.

h	Model	Stellar (XLM)						
		MAE	RMSE	MAPE	RMSPE	LLE	ScMAE	ScRMSE
1	MEM(2,1)	26.2949	46.0738	0.4655	0.6373	0.1271	0.6595	0.7883
	MEM(1,1)	27.1366	46.7577	0.4966	0.6871	0.1308	0.6806	0.8001
	HAR	26.9736	46.8549	0.4887	0.6758	0.1291	0.6765	0.8017
2	MEM(2,1)	28.3225	49.4200	0.5192	0.7102	0.1480	0.7103	0.8456
	MEM(1,1)	29.6677	50.6133	0.5625	0.7856	0.1552	0.7441	0.8660
	HAR	29.0490	49.6042	0.5532	0.7684	0.1501	0.7286	0.8488
3	MEM(2,1)	29.3613	50.6903	0.5459	0.7508	0.1572	0.7364	0.8673
	MEM(1,1)	31.0915	52.0423	0.6095	0.8551	0.1679	0.7798	0.8905
	HAR	30.1631	50.8741	0.5850	0.8177	0.1598	0.7565	0.8705
4	MEM(2,1)	29.7588	51.2006	0.5606	0.7753	0.1632	0.7464	0.8761
	MEM(1,1)	31.6961	52.5930	0.6444	0.9118	0.1777	0.7949	0.8999
	HAR	30.5502	51.2307	0.6043	0.8499	0.1659	0.7662	0.8766
5	MEM(2,1)	30.4281	52.0634	0.5745	0.7949	0.1693	0.7631	0.8908
	MEM(1,1)	32.8585	53.7291	0.6793	0.9586	0.1873	0.8241	0.9193
	HAR	31.3305	52.1613	0.6241	0.8797	0.1723	0.7858	0.8925

Table A.7: Forecast-error measures for the BCH exchange rate under the MEM and HAR models, rounded to four decimal places.

h	Model	Bitcoin Cash (BCH)						
		MAE	RMSE	MAPE	RMSPE	LLE	ScMAE	ScRMSE
1	MEM(2,1)	25.3121	42.3906	0.4955	0.7121	0.1290	0.7079	0.8054
	MEM(1,1)	25.7714	42.4421	0.5182	0.7389	0.1317	0.7207	0.8064
	HAR	25.6069	42.6251	0.5058	0.7276	0.1301	0.7161	0.8099
2	MEM(2,1)	27.0246	44.6715	0.5452	0.7694	0.1477	0.7558	0.8488
	MEM(1,1)	27.7645	44.9334	0.5716	0.8090	0.1502	0.7764	0.8537
	HAR	27.4867	44.8015	0.5648	0.7991	0.1491	0.7687	0.8512
3	MEM(2,1)	28.0859	46.1836	0.5714	0.8062	0.1579	0.7854	0.8775
	MEM(1,1)	28.9791	46.5624	0.6095	0.8649	0.1617	0.8104	0.8847
	HAR	28.6146	46.3000	0.5950	0.8408	0.1592	0.8002	0.8797
4	MEM(2,1)	28.6412	46.8051	0.5889	0.8364	0.1634	0.8010	0.8893
	MEM(1,1)	29.6903	47.2838	0.6390	0.9144	0.1692	0.8303	0.8984
	HAR	29.1772	46.9149	0.6151	0.8757	0.1650	0.8160	0.8914
5	MEM(2,1)	29.2009	47.5974	0.6039	0.8582	0.1685	0.8166	0.9043
	MEM(1,1)	30.4695	48.3049	0.6640	0.9512	0.1760	0.8521	0.9178
	HAR	29.8300	47.8100	0.6342	0.9045	0.1707	0.8342	0.9084

Table A.8: Forecast-error measures for the DOGE exchange rate under the MEM and HAR models, rounded to four decimal places.

h	Model	Dogecoin (DOGE)						
		MAE	RMSE	MAPE	RMSPE	LLE	ScMAE	ScRMSE
1	MEM(2,1)	33.5819	68.6198	0.5223	0.7244	0.1545	0.6438	0.7463
	MEM(1,1)	34.5795	69.9145	0.5589	0.7839	0.1674	0.6629	0.7604
	HAR	34.0563	69.1428	0.5499	0.7695	0.1583	0.6529	0.7520
2	MEM(2,1)	36.8509	75.2572	0.5968	0.8286	0.1965	0.7065	0.8185
	MEM(1,1)	38.9308	78.0037	0.6545	0.9364	0.2171	0.7463	0.8484
	HAR	37.6591	76.1338	0.6396	0.9020	0.2040	0.7220	0.8281
3	MEM(2,1)	38.3915	78.2935	0.6331	0.8845	0.2185	0.7360	0.8516
	MEM(1,1)	41.4908	81.9591	0.7218	1.0380	0.2444	0.7954	0.8914
	HAR	39.4246	79.3339	0.6837	0.9707	0.2268	0.7558	0.8629
4	MEM(2,1)	39.3502	79.7364	0.6539	0.9148	0.2324	0.7544	0.8673
	MEM(1,1)	43.3406	84.0279	0.7760	1.1121	0.2661	0.8309	0.9139
	HAR	40.5812	80.8813	0.7115	1.0094	0.2413	0.7780	0.8797
5	MEM(2,1)	40.1880	80.5450	0.6695	0.9374	0.2418	0.7704	0.8761
	MEM(1,1)	44.7631	85.1387	0.8216	1.1859	0.2805	0.8581	0.9260
	HAR	41.5820	81.8018	0.7343	1.0435	0.2510	0.7972	0.8897

Table A.9: Forecast-error measures for the ETC exchange rate under the MEM and HAR models, rounded to four decimal places.

h	Model	Ethereum Classic (ETC)						
		MAE	RMSE	MAPE	RMSPE	LLE	ScMAE	ScRMSE
1	MEM(2,1)	29.7134	50.3104	0.5024	0.7001	0.1362	0.7222	0.7855
	MEM(1,1)	30.4620	50.9525	0.5310	0.7492	0.1412	0.7404	0.7955
	HAR	30.2253	50.7246	0.5141	0.7151	0.1374	0.7346	0.7920
2	MEM(2,1)	31.6433	53.6196	0.5540	0.7869	0.1598	0.7691	0.8372
	MEM(1,1)	32.8132	54.5292	0.5905	0.8598	0.1656	0.7975	0.8514
	HAR	32.3047	53.7827	0.5703	0.8091	0.1605	0.7852	0.8397
3	MEM(2,1)	32.5422	55.1699	0.5761	0.8285	0.1695	0.7910	0.8614
	MEM(1,1)	33.9774	56.2973	0.6286	0.9346	0.1774	0.8258	0.8790
	HAR	33.3227	55.2505	0.5953	0.8533	0.1693	0.8099	0.8626
4	MEM(2,1)	32.9546	55.5705	0.5883	0.8490	0.1740	0.8010	0.8676
	MEM(1,1)	34.5622	56.8038	0.6567	0.9846	0.1846	0.8400	0.8869
	HAR	33.7127	55.6353	0.6088	0.8777	0.1738	0.8194	0.8687
5	MEM(2,1)	33.5679	56.5023	0.6010	0.8683	0.1794	0.8159	0.8822
	MEM(1,1)	35.5034	58.0806	0.6853	1.0342	0.1932	0.8629	0.9068
	HAR	34.3571	56.5861	0.6240	0.9038	0.1790	0.8351	0.8835

Table A.10: Forecast-error measures for the TRX exchange rate under the MEM and HAR models, rounded to four decimal places.

h	Model	Tron (TRX)						
		MAE	RMSE	MAPE	RMSPE	LLE	ScMAE	ScRMSE
1	MEM(2,1)	23.2602	38.0984	0.5496	0.7607	0.1391	0.5349	0.7028
	MEM(1,1)	23.8712	38.3502	0.5853	0.8102	0.1443	0.5490	0.7075
	HAR	24.0112	38.8151	0.5845	0.8179	0.1453	0.5522	0.7160
2	MEM(2,1)	25.5370	40.7740	0.6332	0.8777	0.1649	0.5873	0.7522
	MEM(1,1)	26.3254	41.2225	0.6725	0.9376	0.1703	0.6054	0.7604
	HAR	26.4347	41.2702	0.6853	0.9636	0.1737	0.6079	0.7613
3	MEM(2,1)	26.9828	42.3115	0.6836	0.9561	0.1777	0.6205	0.7805
	MEM(1,1)	28.2214	43.0597	0.7483	1.0549	0.1866	0.6490	0.7943
	HAR	28.1024	42.8712	0.7445	1.0517	0.1875	0.6463	0.7909
4	MEM(2,1)	27.8508	43.3206	0.7190	1.0133	0.1867	0.6405	0.7992
	MEM(1,1)	29.4185	44.4048	0.8102	1.1554	0.2002	0.6765	0.8192
	HAR	29.0580	43.8458	0.7864	1.1200	0.1975	0.6682	0.8088
5	MEM(2,1)	28.6468	44.2118	0.7483	1.0574	0.1945	0.6588	0.8156
	MEM(1,1)	30.6393	45.6199	0.8655	1.2389	0.2129	0.7046	0.8416
	HAR	30.0874	44.8905	0.8256	1.1772	0.2074	0.6919	0.8281

Table A.11: Forecast-error measures for the LINK exchange rate under the MEM and HAR models, rounded to four decimal places.

h	Model	Chainlink (LINK)						
		MAE	RMSE	MAPE	RMSPE	LLE	ScMAE	ScRMSE
1	MEM(2,1)	27.9794	45.7529	0.4331	0.6202	0.1017	0.6346	0.7706
	MEM(1,1)	28.9431	46.2797	0.4670	0.6781	0.1060	0.6564	0.7795
	HAR	28.4718	45.9992	0.4581	0.6657	0.1050	0.6457	0.7747
2	MEM(2,1)	30.1007	49.1869	0.4881	0.7113	0.1212	0.6827	0.8284
	MEM(1,1)	31.5596	49.8040	0.5329	0.7893	0.1262	0.7158	0.8388
	HAR	30.9106	49.5499	0.5218	0.7719	0.1262	0.7010	0.8345
3	MEM(2,1)	31.3835	50.4010	0.5170	0.7544	0.1278	0.7118	0.8489
	MEM(1,1)	33.2142	51.0761	0.5792	0.8618	0.1348	0.7533	0.8602
	HAR	32.2971	50.8279	0.5533	0.8189	0.1335	0.7325	0.8561
4	MEM(2,1)	32.2113	51.1201	0.5361	0.7810	0.1327	0.7306	0.8610
	MEM(1,1)	34.4407	52.0059	0.6172	0.9156	0.1432	0.7811	0.8759
	HAR	33.2630	51.6132	0.5756	0.8507	0.1389	0.7544	0.8693
5	MEM(2,1)	32.9447	52.3510	0.5532	0.8108	0.1390	0.7472	0.8817
	MEM(1,1)	35.8379	53.8322	0.6530	0.9728	0.1530	0.8128	0.9067
	HAR	34.0614	52.8845	0.5959	0.8865	0.1457	0.7725	0.8907

Table A.12: Forecast-error measures for the BNB exchange rate under the MEM and HAR models, rounded to four decimal places.

h	Model	Binance Coin (BNB)						
		MAE	RMSE	MAPE	RMSPE	LLE	ScMAE	ScRMSE
1	MEM(2,1)	22.6165	37.0886	0.4969	0.6990	0.1205	0.6187	0.7115
	MEM(1,1)	23.0143	37.2237	0.5137	0.7236	0.1226	0.6296	0.7141
	HAR	23.3176	37.9917	0.5112	0.7232	0.1243	0.6379	0.7288
2	MEM(2,1)	24.6886	40.5619	0.5665	0.8097	0.1430	0.6754	0.7781
	MEM(1,1)	25.1186	40.5704	0.5829	0.8365	0.1445	0.6872	0.7783
	HAR	24.8677	40.5657	0.5840	0.8440	0.1458	0.6803	0.7782
3	MEM(2,1)	25.8080	42.4567	0.6066	0.8835	0.1540	0.7061	0.8144
	MEM(1,1)	26.3723	42.6039	0.6335	0.9309	0.1568	0.7215	0.8173
	HAR	26.0712	42.7224	0.6250	0.9156	0.1573	0.7133	0.8195
4	MEM(2,1)	26.7397	43.5157	0.6395	0.9387	0.1630	0.7315	0.8348
	MEM(1,1)	27.4275	43.8196	0.6776	1.0062	0.1677	0.7504	0.8406
	HAR	27.0012	43.5758	0.6589	0.9710	0.1657	0.7387	0.8359
5	MEM(2,1)	27.5067	44.8591	0.6661	0.9825	0.1722	0.7525	0.8605
	MEM(1,1)	28.2027	45.3899	0.7118	1.0643	0.1788	0.7716	0.8707
	HAR	27.7951	45.0917	0.6863	1.0174	0.1757	0.7604	0.8650

Table A.13: Forecast-error measures for the AUD exchange rate under the MEM and HAR models, rounded to four decimal places.

h	Model	Australian dollar (AUD)						
		MAE	RMSE	MAPE	RMSPE	LLE	ScMAE	ScRMSE
1	MEM(2,1)	3.4608	6.1864	0.3744	1.0597	0.0913	0.8642	0.8829
	MEM(1,1)	3.4693	6.2005	0.3749	1.0597	0.0915	0.8663	0.8850
	HAR	3.5221	6.1860	0.3838	1.2159	0.0910	0.8795	0.8829
2	MEM(2,1)	3.4800	6.2645	0.3754	1.0607	0.0940	0.8690	0.8941
	MEM(1,1)	3.4785	6.2404	0.3750	1.0601	0.0930	0.8686	0.8906
	HAR	3.5046	6.2159	0.3774	1.2075	0.0917	0.8751	0.8872
3	MEM(2,1)	3.4887	6.3017	0.3756	1.0993	0.0958	0.8711	0.8994
	MEM(1,1)	3.4868	6.2780	0.3753	1.0982	0.0948	0.8706	0.8960
	HAR	3.5263	6.2730	0.3787	1.2658	0.0939	0.8805	0.8953
4	MEM(2,1)	3.5170	6.3430	0.3790	1.1668	0.0975	0.8782	0.9053
	MEM(1,1)	3.5167	6.3219	0.3787	1.1652	0.0965	0.8781	0.9023
	HAR	3.5453	6.3330	0.3825	1.3571	0.0955	0.8853	0.9039
5	MEM(2,1)	3.5479	6.4038	0.3838	1.2638	0.1000	0.8859	0.9140
	MEM(1,1)	3.5493	6.3871	0.3837	1.2625	0.0990	0.8863	0.9116
	HAR	3.5853	6.4150	0.3910	1.5348	0.0978	0.8952	0.9156

Table A.14: Forecast-error measures for the CAD exchange rate under the MEM and HAR models, rounded to four decimal places.

h	Model	Canadian dollar (CAD)						
		MAE	RMSE	MAPE	RMSPE	LLE	ScMAE	ScRMSE
1	MEM(2,1)	2.1844	3.0995	0.3708	0.4976	0.0836	0.8440	0.8801
	MEM(1,1)	2.1842	3.1006	0.3712	0.4986	0.0837	0.8440	0.8804
	HAR	2.1573	3.0739	0.3674	0.4970	0.0830	0.8336	0.8728
2	MEM(2,1)	2.1935	3.1323	0.3723	0.5010	0.0857	0.8476	0.8894
	MEM(1,1)	2.1902	3.1211	0.3723	0.5009	0.0851	0.8463	0.8862
	HAR	2.1560	3.1040	0.3659	0.4968	0.0843	0.8331	0.8813
3	MEM(2,1)	2.2176	3.1677	0.3760	0.5054	0.0878	0.8569	0.8994
	MEM(1,1)	2.2145	3.1573	0.3760	0.5055	0.0872	0.8557	0.8965
	HAR	2.1884	3.1481	0.3711	0.5032	0.0865	0.8456	0.8939
4	MEM(2,1)	2.2255	3.1796	0.3776	0.5079	0.0885	0.8599	0.9028
	MEM(1,1)	2.2231	3.1706	0.3778	0.5082	0.0880	0.8590	0.9003
	HAR	2.2019	3.1614	0.3742	0.5074	0.0870	0.8508	0.8976
5	MEM(2,1)	2.2430	3.2139	0.3805	0.5121	0.0909	0.8667	0.9125
	MEM(1,1)	2.2418	3.2062	0.3808	0.5125	0.0904	0.8662	0.9104
	HAR	2.2267	3.2021	0.3789	0.5137	0.0894	0.8604	0.9092

Table A.15: Forecast-error measures for the EUR exchange rate under the MEM and HAR models, rounded to four decimal places.

h	Model	Euro (EUR)						
		MAE	RMSE	MAPE	RMSPE	LLE	ScMAE	ScRMSE
1	MEM(2,1)	2.1871	3.0148	0.3452	0.4567	0.0751	0.7549	0.7832
	MEM(1,1)	2.1868	3.0129	0.3453	0.4572	0.0751	0.7548	0.7827
	HAR	2.2109	3.0058	0.3545	0.4707	0.0753	0.7631	0.7808
2	MEM(2,1)	2.2081	3.0516	0.3479	0.4600	0.0766	0.7622	0.7927
	MEM(1,1)	2.2054	3.0448	0.3478	0.4600	0.0763	0.7612	0.7910
	HAR	2.2413	3.0526	0.3588	0.4757	0.0770	0.7736	0.7930
3	MEM(2,1)	2.2200	3.0699	0.3496	0.4626	0.0773	0.7663	0.7975
	MEM(1,1)	2.2172	3.0637	0.3495	0.4627	0.0770	0.7653	0.7959
	HAR	2.2613	3.0740	0.3622	0.4804	0.0778	0.7805	0.7986
4	MEM(2,1)	2.2340	3.0993	0.3513	0.4641	0.0784	0.7711	0.8051
	MEM(1,1)	2.2320	3.0934	0.3513	0.4643	0.0781	0.7704	0.8036
	HAR	2.2759	3.1101	0.3643	0.4825	0.0789	0.7856	0.8079
5	MEM(2,1)	2.2509	3.1338	0.3543	0.4689	0.0802	0.7770	0.8141
	MEM(1,1)	2.2494	3.1281	0.3543	0.4690	0.0799	0.7764	0.8126
	HAR	2.2947	3.1525	0.3684	0.4896	0.0808	0.7921	0.8190

Table A.16: Forecast-error measures for the GBP exchange rate under the MEM and HAR models, rounded to four decimal places.

h	Model	Pound sterling (GBP)					LLE	ScMAE	ScRMSE
		MAE	RMSE	MAPE	RMSPE				
1	MEM(2,1)	2.8266	4.1274	0.3558	0.4992	0.0798	0.8402	0.8257	
	MEM(1,1)	2.8258	4.1222	0.3555	0.4985	0.0795	0.8399	0.8247	
	HAR	2.8724	4.1625	0.3597	0.4993	0.0815	0.8538	0.8327	
2	MEM(2,1)	2.8620	4.2187	0.3610	0.5110	0.0826	0.8507	0.8440	
	MEM(1,1)	2.8617	4.2209	0.3607	0.5102	0.0825	0.8506	0.8444	
	HAR	2.8979	4.2735	0.3658	0.5165	0.0850	0.8614	0.8549	
3	MEM(2,1)	2.9050	4.3083	0.3669	0.5203	0.0856	0.8635	0.8619	
	MEM(1,1)	2.9041	4.3077	0.3666	0.5198	0.0854	0.8632	0.8618	
	HAR	2.9419	4.3593	0.3719	0.5248	0.0883	0.8745	0.8721	
4	MEM(2,1)	2.9346	4.3915	0.3711	0.5336	0.0880	0.8723	0.8785	
	MEM(1,1)	2.9302	4.3874	0.3705	0.5328	0.0877	0.8710	0.8777	
	HAR	2.9791	4.4363	0.3765	0.5373	0.0904	0.8855	0.8875	
5	MEM(2,1)	2.9997	4.4762	0.3792	0.5437	0.0912	0.8916	0.8955	
	MEM(1,1)	2.9925	4.4688	0.3783	0.5428	0.0908	0.8895	0.8940	
	HAR	3.0399	4.5210	0.3847	0.5482	0.0937	0.9036	0.9044	

Table A.17: Forecast-error measures for the JPY exchange rate under the MEM and HAR models, rounded to four decimal places.

h	Model	Japanese yen (JPY)					LLE	ScMAE	ScRMSE
		MAE	RMSE	MAPE	RMSPE				
1	MEM(2,1)	2.7838	4.4389	0.4528	0.7387	0.1180	0.7565	0.8072	
	MEM(1,1)	2.7780	4.4266	0.4525	0.7386	0.1178	0.7550	0.8050	
	HAR	2.8013	4.4637	0.4557	0.7461	0.1194	0.7613	0.8117	
2	MEM(2,1)	2.8807	4.5682	0.4695	0.7585	0.1257	0.7829	0.8307	
	MEM(1,1)	2.8541	4.5343	0.4648	0.7535	0.1243	0.7756	0.8245	
	HAR	2.9134	4.5843	0.4772	0.7703	0.1267	0.7917	0.8336	
3	MEM(2,1)	2.9288	4.6368	0.4776	0.7699	0.1302	0.7959	0.8432	
	MEM(1,1)	2.9082	4.6067	0.4740	0.7672	0.1291	0.7903	0.8377	
	HAR	2.9650	4.6508	0.4880	0.7857	0.1311	0.8058	0.8457	
4	MEM(2,1)	2.9621	4.7146	0.4820	0.7817	0.1348	0.8050	0.8573	
	MEM(1,1)	2.9475	4.6903	0.4797	0.7827	0.1339	0.8010	0.8529	
	HAR	3.0006	4.7338	0.4942	0.8018	0.1358	0.8154	0.8608	
5	MEM(2,1)	2.9889	4.7511	0.4848	0.7898	0.1368	0.8123	0.8640	
	MEM(1,1)	2.9764	4.7332	0.4830	0.7939	0.1361	0.8089	0.8607	
	HAR	3.0313	4.7565	0.4989	0.8117	0.1371	0.8238	0.8649	

Table A.18: Forecast-error measures for the BRL exchange rate under the MEM and HAR models, rounded to four decimal places.

h	Model	Brazilian real (BRL)				LLE	ScMAE	ScRMSE
		MAE	RMSE	MAPE	RMSPE			
1	MEM(2,1)	4.5509	6.3091	0.6714	2.7121	0.1098	0.8694	0.8950
	MEM(1,1)	4.5489	6.3089	0.6741	2.7429	0.1097	0.8690	0.8950
	HAR	4.5443	6.3260	0.6701	2.7316	0.1101	0.8681	0.8974
2	MEM(2,1)	4.6055	6.3524	0.6843	2.7926	0.1106	0.8798	0.9012
	MEM(1,1)	4.6060	6.3608	0.6839	2.7919	0.1109	0.8799	0.9024
	HAR	4.6005	6.3633	0.6853	2.8383	0.1107	0.8789	0.9027
3	MEM(2,1)	4.6317	6.3781	0.6811	2.7093	0.1114	0.8848	0.9048
	MEM(1,1)	4.6330	6.3869	0.6800	2.7005	0.1117	0.8851	0.9061
	HAR	4.6265	6.3926	0.6810	2.7276	0.1116	0.8838	0.9069
4	MEM(2,1)	4.6291	6.3936	0.6800	2.7252	0.1115	0.8843	0.9070
	MEM(1,1)	4.6304	6.4022	0.6787	2.7164	0.1118	0.8846	0.9082
	HAR	4.6231	6.4080	0.6802	2.7529	0.1118	0.8832	0.9091
5	MEM(2,1)	4.6533	6.4193	0.6811	2.7050	0.1125	0.8890	0.9107
	MEM(1,1)	4.6600	6.4302	0.6799	2.6942	0.1128	0.8903	0.9122
	HAR	4.6465	6.4336	0.6802	2.7099	0.1128	0.8877	0.9127

Table A.19: Forecast-error measures for the CHF exchange rate under the MEM and HAR models, rounded to four decimal places.

h	Model	Swiss franc (CHF)				LLE	ScMAE	ScRMSE
		MAE	RMSE	MAPE	RMSPE			
1	MEM(2,1)	2.2792	3.1486	0.3551	0.4897	0.0739	0.8359	0.8358
	MEM(1,1)	2.2779	3.1463	0.3546	0.4894	0.0740	0.8354	0.8352
	HAR	2.3658	3.2456	0.3755	0.5173	0.0801	0.8677	0.8616
2	MEM(2,1)	2.3100	3.1800	0.3599	0.4925	0.0757	0.8472	0.8442
	MEM(1,1)	2.3050	3.1792	0.3580	0.4912	0.0758	0.8454	0.8440
	HAR	2.4048	3.2858	0.3840	0.5281	0.0823	0.8820	0.8723
3	MEM(2,1)	2.3238	3.2188	0.3612	0.4934	0.0777	0.8523	0.8545
	MEM(1,1)	2.3226	3.2197	0.3600	0.4923	0.0780	0.8518	0.8547
	HAR	2.4313	3.3313	0.3885	0.5340	0.0846	0.8917	0.8843
4	MEM(2,1)	2.3299	3.2390	0.3610	0.4885	0.0787	0.8545	0.8598
	MEM(1,1)	2.3286	3.2426	0.3600	0.4875	0.0791	0.8540	0.8608
	HAR	2.4485	3.3510	0.3914	0.5334	0.0859	0.8980	0.8896
5	MEM(2,1)	2.3432	3.2845	0.3612	0.4891	0.0807	0.8594	0.8719
	MEM(1,1)	2.3455	3.2936	0.3607	0.4888	0.0813	0.8602	0.8743
	HAR	2.4774	3.4132	0.3956	0.5414	0.0888	0.9086	0.9061

Table A.20: Forecast-error measures for the KRW exchange rate under the MEM and HAR models, rounded to four decimal places.

h	Model	South Korean won (KRW)						
		MAE	RMSE	MAPE	RMSPE	LLE	ScMAE	ScRMSE
1	MEM(2,1)	2.8965	4.4242	0.3450	0.4842	0.0807	0.8397	0.8895
	MEM(1,1)	2.8998	4.4343	0.3453	0.4844	0.0810	0.8407	0.8916
	HAR	2.9137	4.3889	0.3519	0.4971	0.0796	0.8447	0.8825
2	MEM(2,1)	2.9215	4.4797	0.3475	0.4898	0.0827	0.8469	0.9007
	MEM(1,1)	2.9195	4.4770	0.3468	0.4883	0.0826	0.8464	0.9002
	HAR	2.9254	4.4447	0.3500	0.4968	0.0813	0.8481	0.8937
3	MEM(2,1)	2.9405	4.5183	0.3489	0.4922	0.0843	0.8525	0.9085
	MEM(1,1)	2.9401	4.5175	0.3484	0.4909	0.0843	0.8523	0.9083
	HAR	2.9577	4.4930	0.3535	0.5026	0.0832	0.8575	0.9034
4	MEM(2,1)	2.9495	4.5399	0.3485	0.4926	0.0851	0.8551	0.9128
	MEM(1,1)	2.9507	4.5387	0.3482	0.4916	0.0850	0.8554	0.9126
	HAR	2.9762	4.5188	0.3553	0.5065	0.0841	0.8628	0.9086
5	MEM(2,1)	2.9620	4.5601	0.3492	0.4942	0.0859	0.8587	0.9169
	MEM(1,1)	2.9632	4.5589	0.3490	0.4936	0.0857	0.8590	0.9166
	HAR	2.9963	4.5461	0.3578	0.5126	0.0849	0.8686	0.9141

Table A.21: Forecast-error measures for the TRY exchange rate under the MEM and HAR models, rounded to four decimal places.

h	Model	Turkish lira (TRY)						
		MAE	RMSE	MAPE	RMSPE	LLE	ScMAE	ScRMSE
1	MEM(2,1)	5.7332	12.0405	0.6499	1.2906	0.1849	0.6101	0.7112
	MEM(1,1)	5.7717	11.8656	0.6821	1.3386	0.1886	0.6142	0.7009
	HAR	5.8306	12.0637	0.7060	1.3709	0.1862	0.6205	0.7126
2	MEM(2,1)	6.2620	12.8342	0.7556	1.3729	0.2174	0.6664	0.7581
	MEM(1,1)	6.3155	12.6467	0.7857	1.4622	0.2210	0.6721	0.7470
	HAR	6.3879	12.7532	0.8288	1.4896	0.2204	0.6798	0.7533
3	MEM(2,1)	6.5401	13.6359	0.7975	1.4176	0.2305	0.6960	0.8055
	MEM(1,1)	6.6554	13.5103	0.8443	1.5196	0.2356	0.7082	0.7981
	HAR	6.6654	13.5350	0.8799	1.5526	0.2356	0.7093	0.7995
4	MEM(2,1)	6.8575	14.1591	0.8419	1.4302	0.2432	0.7297	0.8364
	MEM(1,1)	7.0089	14.0890	0.9070	1.5493	0.2491	0.7459	0.8322
	HAR	6.9805	14.0388	0.9273	1.5705	0.2491	0.7428	0.8293
5	MEM(2,1)	7.0573	14.5734	0.8846	1.5497	0.2520	0.7510	0.8609
	MEM(1,1)	7.1484	14.5099	0.9652	1.7000	0.2600	0.7607	0.8571
	HAR	7.1984	14.4595	0.9747	1.6933	0.2593	0.7660	0.8541

Table A.22: Forecast-error measures for the TWD exchange rate under the MEM and HAR models, rounded to four decimal places.

h	Model	New Taiwan dollar (TWD)						
		MAE	RMSE	MAPE	RMSPE	LLE	ScMAE	ScRMSE
1	MEM(2,1)	7.5997	15.6859	0.9016	1.4894	0.3406	1.0039	0.9255
	MEM(1,1)	7.6097	15.6261	0.8978	1.4447	0.3439	1.0052	0.9220
	HAR	7.1160	15.8602	0.7808	1.2420	0.3682	0.9400	0.9358
2	MEM(2,1)	7.7939	15.7403	0.9300	1.4678	0.3573	1.0295	0.9288
	MEM(1,1)	7.6639	15.7362	0.9016	1.4428	0.3573	1.0123	0.9285
	HAR	7.1010	15.8774	0.7776	1.2176	0.3826	0.9380	0.9368
3	MEM(2,1)	7.7921	15.7719	0.9318	1.4753	0.3577	1.0293	0.9306
	MEM(1,1)	7.6493	15.7802	0.8994	1.4472	0.3584	1.0104	0.9311
	HAR	7.0845	15.9397	0.7747	1.2284	0.3852	0.9358	0.9405
4	MEM(2,1)	7.8148	15.8555	0.9315	1.4677	0.3648	1.0323	0.9356
	MEM(1,1)	7.6593	15.8557	0.8967	1.4342	0.3650	1.0117	0.9356
	HAR	7.0740	16.0219	0.7684	1.2022	0.3929	0.9344	0.9454
5	MEM(2,1)	7.7766	15.8351	0.9256	1.4565	0.3640	1.0272	0.9343
	MEM(1,1)	7.6060	15.8221	0.8887	1.4186	0.3641	1.0047	0.9336
	HAR	7.0016	15.9837	0.7569	1.1672	0.3914	0.9249	0.9431

Table A.23: Forecast-error measures for the RUB exchange rate under the MEM and HAR models, rounded to four decimal places.

h	Model	Russian ruble (RUB)						
		MAE	RMSE	MAPE	RMSPE	LLE	ScMAE	ScRMSE
1	MEM(2,1)	6.9608	15.7424	1.1978	25.7343	0.0980	0.5608	0.5769
	MEM(1,1)	6.9369	15.6242	1.1973	25.6208	0.0979	0.5589	0.5726
	HAR	6.9041	15.5535	1.1721	25.4085	0.0983	0.5562	0.5700
2	MEM(2,1)	7.5128	17.5051	1.1663	22.7876	0.1136	0.6053	0.6415
	MEM(1,1)	7.4128	17.0789	1.1748	23.1536	0.1110	0.5972	0.6259
	HAR	7.4887	17.5141	1.1547	22.3521	0.1135	0.6033	0.6419
3	MEM(2,1)	7.8831	18.4675	1.1128	19.9689	0.1262	0.6351	0.6768
	MEM(1,1)	7.8101	18.0801	1.1065	19.7154	0.1236	0.6292	0.6626
	HAR	7.8808	18.4376	1.1013	19.1518	0.1258	0.6349	0.6757
4	MEM(2,1)	8.1981	19.3668	1.1263	20.4824	0.1396	0.6605	0.7098
	MEM(1,1)	8.0977	19.0378	1.1350	20.6265	0.1376	0.6524	0.6977
	HAR	8.1749	19.3594	1.1274	19.8158	0.1394	0.6586	0.7095
5	MEM(2,1)	8.4519	19.9005	1.1298	20.8293	0.1498	0.6809	0.7293
	MEM(1,1)	8.4181	19.5643	1.1580	21.4411	0.1493	0.6782	0.7170
	HAR	8.4672	19.8900	1.1373	20.3113	0.1506	0.6821	0.7289

Table A.24: Forecast-error measures for the INR exchange rate under the MEM and HAR models, rounded to four decimal places.

h	Model	Indian rupee (INR)				LLE	ScMAE	ScRMSE
		MAE	RMSE	MAPE	RMSPE			
1	MEM(2,1)	2.3798	3.5177	0.4464	0.7139	0.1118	0.7390	0.7515
	MEM(1,1)	2.3476	3.4662	0.4445	0.7074	0.1112	0.7290	0.7405
	HAR	2.3674	3.5802	0.4293	0.7096	0.1160	0.7352	0.7648
2	MEM(2,1)	2.4627	3.6371	0.4637	0.7089	0.1153	0.7648	0.7770
	MEM(1,1)	2.4120	3.5498	0.4576	0.7161	0.1143	0.7490	0.7584
	HAR	2.4430	3.6846	0.4481	0.7064	0.1173	0.7587	0.7872
3	MEM(2,1)	2.5268	3.7182	0.4760	0.7176	0.1188	0.7847	0.7943
	MEM(1,1)	2.4706	3.6302	0.4691	0.7240	0.1178	0.7672	0.7755
	HAR	2.5098	3.7609	0.4609	0.7116	0.1201	0.7794	0.8035
4	MEM(2,1)	2.5655	3.7700	0.4817	0.7179	0.1210	0.7967	0.8054
	MEM(1,1)	2.5121	3.6803	0.4745	0.7195	0.1197	0.7801	0.7862
	HAR	2.5433	3.8093	0.4654	0.7088	0.1217	0.7898	0.8138
5	MEM(2,1)	2.5988	3.8209	0.4870	0.7177	0.1231	0.8070	0.8163
	MEM(1,1)	2.5347	3.7289	0.4779	0.7169	0.1216	0.7871	0.7966
	HAR	2.5737	3.8567	0.4694	0.6969	0.1234	0.7992	0.8239

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